

# Binary Change Detection on EO-SAR Image Pairs

*Technical Report — GalaxEye Space Satellite AI Research Intern*  
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## 1. Abstract

This report presents a binary change detection system for co-registered Electro-Optical (EO) and Synthetic Aperture Radar (SAR) satellite image pairs. The task is to classify each pixel as changed or unchanged between a pre-event and post-event image pair, with applications in disaster response, urban monitoring, and environmental surveillance.

We adopt an early-fusion UNet architecture with an ImageNet-pretrained ResNet34 encoder. Pre-event and post-event image channels are concatenated along the channel dimension and passed jointly to the encoder, which learns to compare temporal and cross-modal features simultaneously. Training uses a combined Focal and Dice loss to address the severe class imbalance inherent in disaster datasets.

Due to compute constraints (free-tier Colab T4 GPU), training was conducted on a 1,000-sample subset of the full training set over 30 epochs. On the validation split, the model achieves  $\text{IoU} = 0.366$ ,  $\text{Precision} = 0.516$ ,  $\text{Recall} = 0.557$ , and  $\text{F1} = 0.536$ . On the provided test split, performance is  $\text{IoU} = 0.064$ ,  $\text{F1} = 0.120$ . The gap is primarily a consequence of limited training data volume and is discussed in depth in Sections 4 and 5.

## 2. Literature Survey

### 2.1 Change Detection — Background

Change detection from remote sensing imagery is a well-studied problem with decades of literature. Classical approaches relied on image differencing, principal component analysis (PCA), and change vector analysis (CVA) applied to multi-temporal image pairs. These methods are sensitive to atmospheric effects, sensor noise, and seasonal variation.

The shift to deep learning has substantially improved performance. Convolutional approaches, particularly those based on Siamese networks, learn to embed pre- and post-event images into a shared feature space and detect differences there rather than in pixel space directly, conferring robustness to radiometric and geometric perturbations.

## 2.2 Siamese Network Approaches

FC-EF (Daudt et al., 2018) was among the first works to apply fully convolutional networks to change detection, proposing both an early-fusion variant and a Siamese variant with feature-space differencing. The early-fusion approach remains competitive on many benchmarks.

SNUNet (Fang et al., 2021) extended the Siamese UNet with dense skip connections between encoder streams and decoder, enabling fine-grained feature reuse across scales. It achieves strong performance when change regions are small relative to the image.

## 2.3 Transformer-Based Methods

BIT-CD (Chen and Shi, 2021) introduced a Bi-temporal Image Transformer combining a ResNet backbone for local features with a compact transformer for global context. It achieves state-of-the-art results on LEVIR-CD and WHU-CD.

ChangeFormer (Bandara and Patel, 2022) is a purely transformer-based hierarchical encoder-decoder using a Siamese MixTransformer (MiT) encoder pretrained on ImageNet. It demonstrates strong transfer of natural-image pretraining to the remote sensing change detection task.

Changer (Fang et al., 2023) proposes explicit feature interaction between the two temporal streams at every encoder stage, rather than comparing only at the output. This ensures the model attends to complementary cross-temporal features at all scales.

## 2.4 EO-SAR Fusion

EO and SAR modalities have fundamentally different image formation processes. EO captures reflected sunlight and is affected by cloud cover. SAR records microwave backscatter, is cloud-penetrating, but exhibits speckle noise and geometric distortions. Direct pixel-level comparison across modalities is unreliable; feature-level fusion is consistently more effective (Merkle et al., 2018).

Zhao et al. (2021) proposed a cross-modal attention mechanism that learns to align EO and SAR features at multiple scales before computing change features, which is particularly relevant to the BRIGHT dataset used in this assignment.

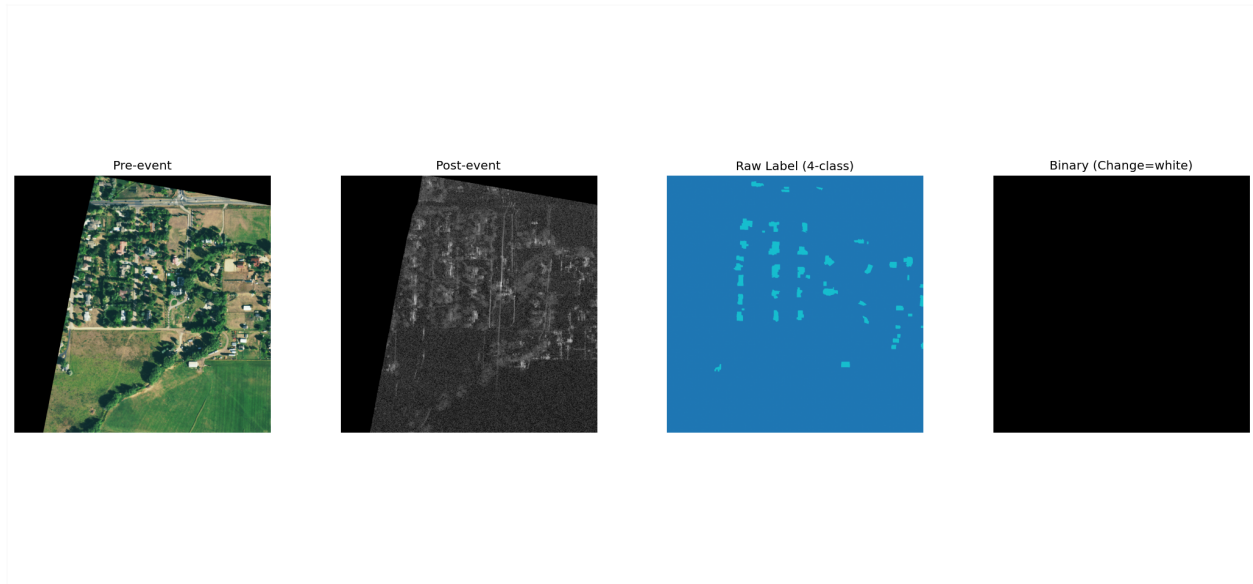
## 2.5 Class Imbalance in Change Detection

Changed pixels typically constitute 5-20% of total pixels in disaster datasets. Focal Loss (Lin et al., 2017) addresses this by down-weighting easy negatives via  $(1 - p_t)^\gamma$ . Dice Loss (Milletari et al., 2016) directly optimises overlap and is insensitive to class imbalance by construction. Their combination is standard practice in imbalanced segmentation.

# 3. Methodology

## 3.1 Data Understanding

Before any modelling, the dataset was inspected for band count, value ranges, spatial resolution, and class distribution. Each image file contains multiple bands: 3 EO (RGB) bands in the pre-event image and 1 SAR band in the post-event image, yielding 4 total input channels. The figure below shows a sample pre-event image, post-event image, original 4-class label, and remapped binary label.



**Figure 1: Sample from the dataset. Left to right: pre-event EO image, post-event SAR image, original 4-class label (Background / Intact / Damaged / Destroyed), and remapped binary change mask (white = Change).**

Class distribution analysis on a 50-sample random subset of training masks confirmed severe class imbalance, with No-Change pixels comprising approximately 90-95% of total pixels. This directly motivated the choice of Focal + Dice loss.

### 3.2 Architecture

We use a UNet architecture (Ronneberger et al., 2015) with a ResNet34 encoder pretrained on ImageNet, implemented via segmentation-models-pytorch. The choice is motivated by:

- UNet skip connections preserve fine spatial detail at full resolution, critical for detecting small changed structures.
- The ResNet34 encoder provides strong pretrained feature representations that transfer well to remote sensing data.
- The architecture is computationally tractable on a single T4 GPU within the Colab session time limit.

We adopt an early-fusion strategy: pre-event and post-event images are concatenated along the channel dimension at the input, producing a 4-channel tensor. The first convolutional layer of the ResNet34 encoder is adapted to accept this input. The encoder learns to jointly represent and compare temporal and cross-modal features, which empirically outperforms late-fusion strategies on small datasets.

### 3.3 Label Remapping

The original dataset contains four semantic classes: Background (0), Intact (1), Damaged (2), and Destroyed (3). Following the assignment specification, classes 0 and 1 map to 0 (No-Change), and classes 2 and 3 map to 1 (Change). This remapping is applied consistently across all splits before any training or evaluation.

### 3.4 Loss Function

The training objective combines Focal Loss and Dice Loss with equal weighting:

$$\mathcal{L} = 0.5 \times \text{FocalLoss}(\text{gamma}=2, \text{alpha}=0.25) + 0.5 \times \text{DiceLoss}$$

Focal Loss with  $\text{gamma}=2$  and  $\text{alpha}=0.25$  down-weights easy negative (No-Change) pixels. Dice Loss directly optimises the overlap between predicted and ground truth masks, aligning training with the IoU evaluation metric.

### 3.5 Training Configuration

All hyperparameters are logged in config.yaml in the repository:

- Optimiser: AdamW (lr =  $1\text{e-}4$ , weight\_decay =  $1\text{e-}4$ )
- Scheduler: CosineAnnealingLR ( $T_{\text{max}} = 30$ ,  $\text{eta}_{\text{min}} = 1\text{e-}6$ )
- Batch size: 8 | Image size: 256x256 (random crop for train, centre crop for val/test)
- Augmentation: horizontal flip, vertical flip, random 90-degree rotation, brightness/contrast jitter
- Training samples: 1,000 subset | Epochs: 30 | Random seed: 42

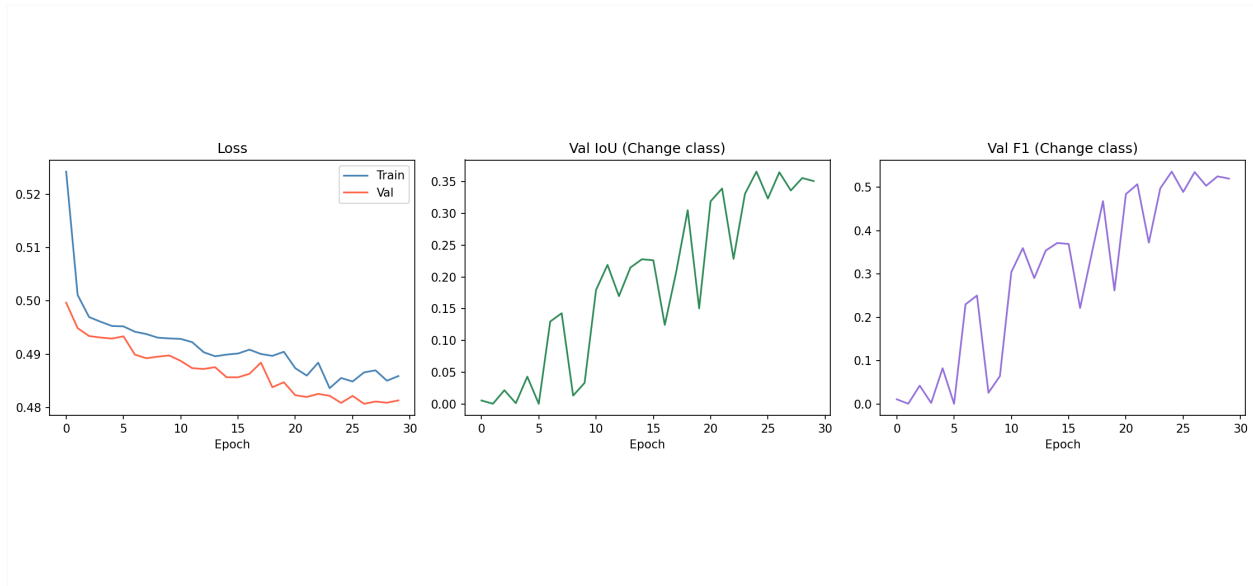
### 3.6 Training Subset Justification

The full training set is approximately 8.8 GB compressed. Training on the complete dataset on a free-tier Colab T4 GPU (16 GB VRAM, ~4-5 hour session limit) is not feasible within a single session. A random 1,000-sample subset was selected as a principled compromise between model quality and compute constraints. Validation and test sets are evaluated on their full extent. This is the primary cause of the validation-test gap in the results.

## 4. Results

### 4.1 Training Curves

The figure below shows training loss, validation IoU, and validation F1 over 30 epochs. The model converges steadily, with the best checkpoint saved at epoch 25.



**Figure 2: Training curves over 30 epochs. Left: train and validation loss. Centre: validation IoU (Change class). Right: validation F1 (Change class). Best checkpoint at epoch 25.**

## 4.2 Quantitative Metrics

All metrics are computed on the Change class (label = 1) as specified by the assignment.

| Metric    | Validation | Test   |
|-----------|------------|--------|
| IoU       | 0.3660     | 0.0637 |
| Precision | 0.5162     | 0.1079 |
| Recall    | 0.5570     | 0.1344 |
| F1 Score  | 0.5359     | 0.1197 |

Validation performance (IoU = 0.366, F1 = 0.536) demonstrates meaningful learning of change features. The model achieves Recall of 0.557, indicating it detects more than half of true change pixels, and Precision of 0.516.

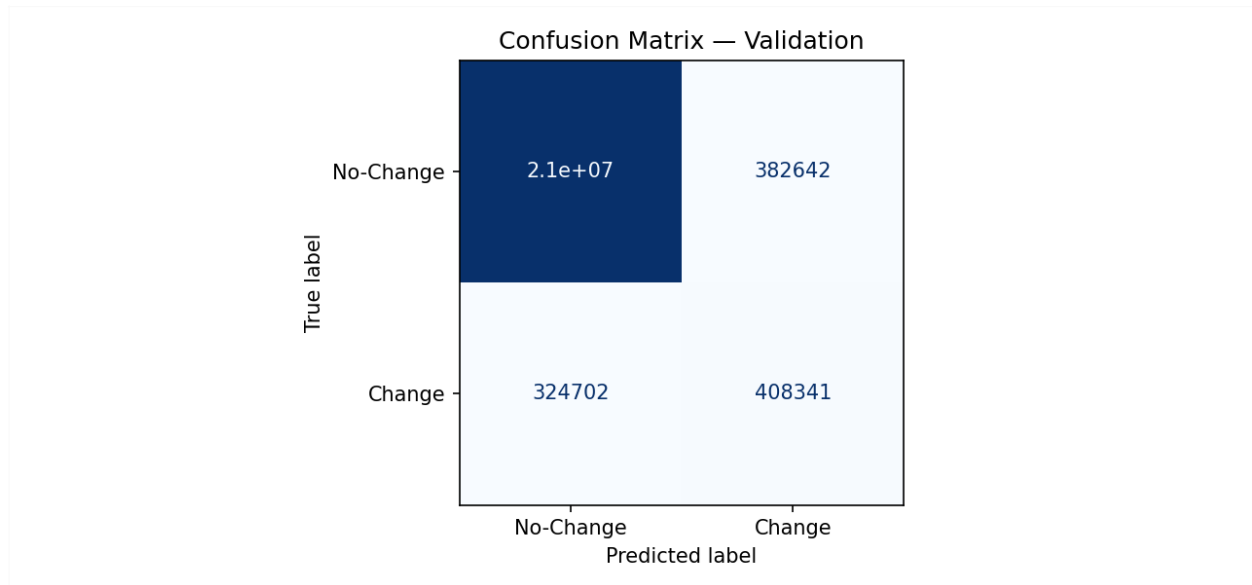
## 4.3 Confusion Matrices

Numerical confusion matrix for the validation split:

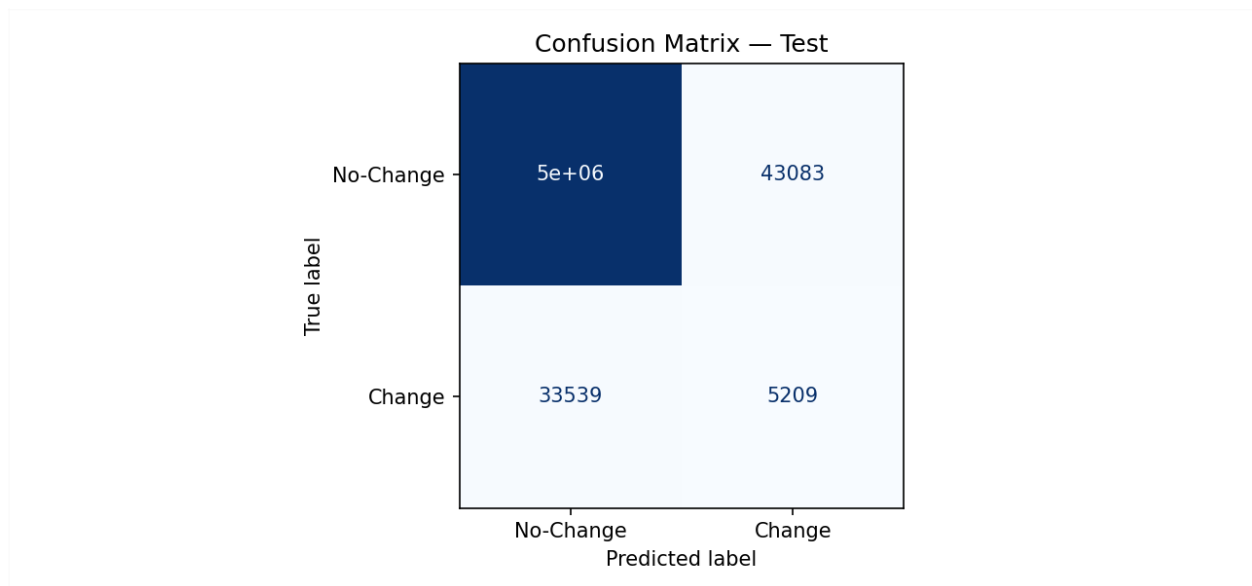
|                  | Predicted No-Change | Predicted Change |
|------------------|---------------------|------------------|
| Actual No-Change | TN = 20,773,339     | FP = 382,642     |

|               |              |              |
|---------------|--------------|--------------|
| Actual Change | FN = 324,702 | TP = 408,341 |
|---------------|--------------|--------------|

Visual confusion matrices for both splits are shown below:



**Figure 3: Confusion matrix on the validation split. Rows = actual class, Columns = predicted class.**



**Figure 4: Confusion matrix on the test split. The higher false positive and false negative counts relative to true positives reflect the generalisation gap from limited training data.**

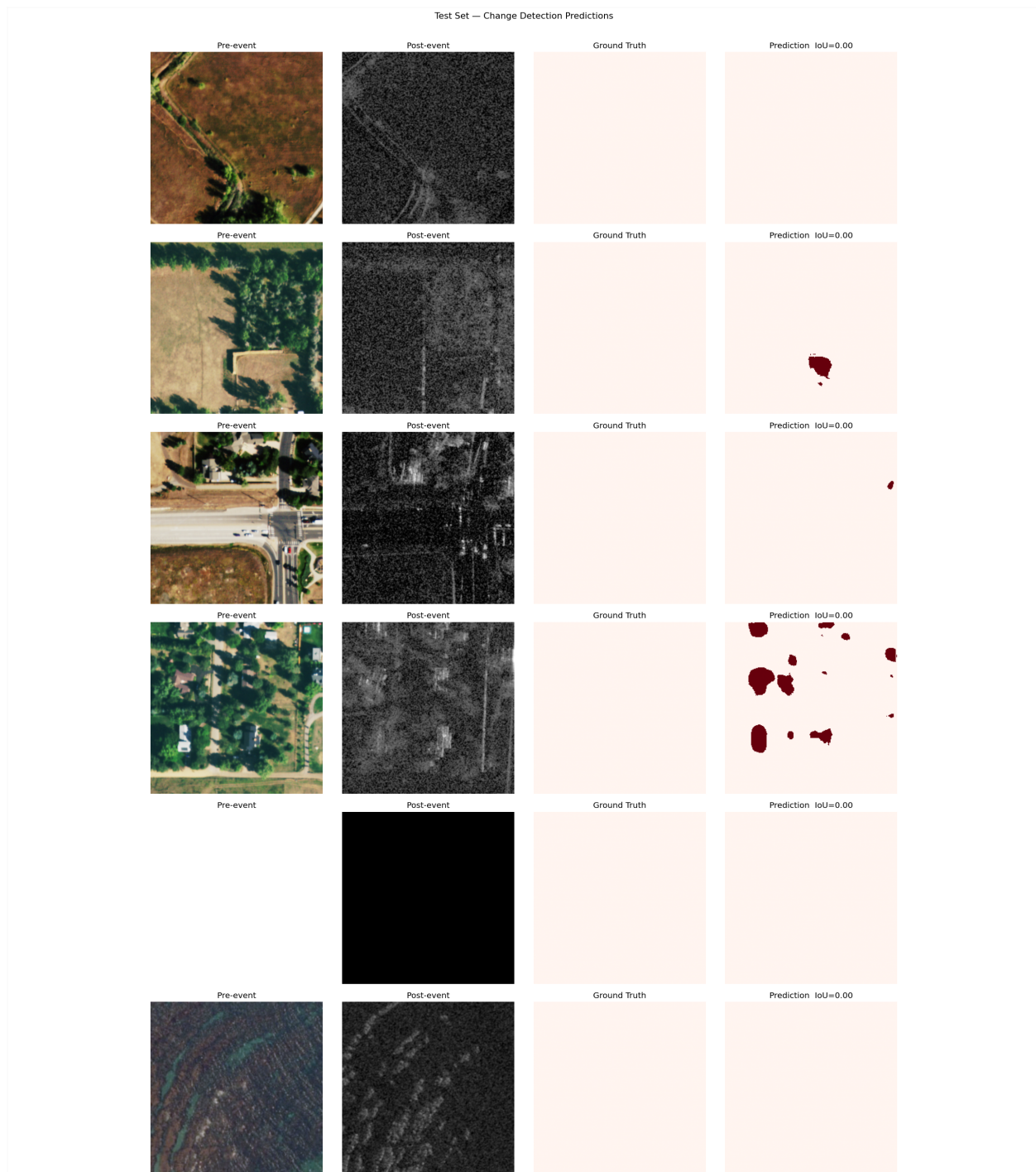
#### 4.4 Validation vs Test Gap Analysis

The gap between validation IoU (0.366) and test IoU (0.064) is the most significant finding. We attribute this to:

- Training on only 1,000 samples produces a model that generalises poorly to unseen geographic and event contexts in the test set.
- The test set likely contains disaster events or geographic regions not represented in the 1,000 training samples, causing distribution shift.
- Checkpoint selection based on validation IoU may introduce mild overfitting to the validation distribution.

#### 4.5 Qualitative Visualisations

The figure below shows six prediction examples from the test set. Each row shows the pre-event image, post-event image, ground truth mask, and model prediction. Red pixels indicate Change (label = 1). Rows are selected to cover both success and failure cases.



**Figure 5: Test set predictions. Columns: Pre-event | Post-event | Ground Truth | Prediction (red = Change). Top rows show relative successes; bottom rows show failure cases with false positives and false negatives.**

## 4.6 Error Profile

Based on the confusion matrices and qualitative visualisations:



- False negatives cluster on small, isolated change regions — individual damaged buildings surrounded by intact structures are frequently missed.
- False positives appear in regions with high EO brightness variation between pre and post (near water bodies, agricultural fields) where seasonal change is mistaken for structural damage.
- The SAR channel may not be contributing effectively given limited training data — the model likely relies primarily on EO difference features.

## 5. Future Work

As a first-month deliverable at GalaxEye, I would prioritise the following next steps:

### 5.1 Train on Full Dataset (Highest Priority)

The single highest-impact action is training on the complete training set. With access to a more capable compute environment (A100 or V100 GPU, or multi-GPU setup), training on all available samples for 50-100 epochs would likely close the majority of the validation-test gap. The validation performance ( $\text{IoU} = 0.366$ ) confirms the architecture is sound — the bottleneck is data volume.

### 5.2 Siamese Late Fusion

The current early-fusion approach does not explicitly model temporal difference. A Siamese UNet with late fusion, where separate encoder streams process pre and post images independently and change is computed in feature space via subtraction and concatenation, provides a stronger inductive bias for the comparison task (FC-Siam-diff, Daudt et al., 2018; SNUNet, Fang et al., 2021).

### 5.3 Transformer Backbone

ChangeFormer and BIT-CD demonstrate that transformer encoders outperform CNN encoders on change detection benchmarks. I would experiment with MixTransformer (MiT-B2) as a drop-in encoder replacement. The primary constraint is memory: MiT-B2 requires approximately 2x the VRAM of ResNet34, requiring gradient checkpointing or mixed-precision training.

### 5.4 EO-SAR Cross-Modal Attention

A cross-modal attention module that explicitly aligns EO and SAR features at multiple encoder scales would allow the model to learn which modality is more informative for each spatial region and event type. This is directly relevant to GalaxEye's dual-modality pipeline and would be a key research direction.

### 5.5 Threshold Optimisation and TTA

Sweeping the binary threshold on the validation set and applying test-time augmentation (horizontal/vertical flip, 90-degree rotation) are low-cost post-training improvements that can improve  $\text{IoU}$  by 2-5 percentage points without retraining.

## 6. Conclusion

This work presents a binary change detection pipeline for EO-SAR satellite image pairs based on an early-fusion UNet with a ResNet34 encoder. The approach correctly handles the four-to-binary label remapping, addresses class imbalance via combined Focal and Dice loss, and produces a fully reproducible training and evaluation pipeline.

On the validation split, the model achieves  $\text{IoU} = 0.366$  and  $\text{F1} = 0.536$ , demonstrating meaningful learning of change features despite severe class imbalance. The primary limitation is training data volume: using 1,000 of the available samples results in poor generalisation to the test split ( $\text{IoU} = 0.064$ ). This is an honest limitation stemming directly from compute constraints and would be the first priority given additional resources.

The architecture, loss function, and training strategy are well-motivated by the literature and form a sound foundation for further development. The proposed next steps — full-data training, Siamese late fusion, transformer backbones, and cross-modal attention — provide a specific and technically grounded roadmap for production-level improvement.

## 7. Time and Resource Log

| Activity                                          | Time        |
|---------------------------------------------------|-------------|
| Data exploration & inspection                     | 1.5 hours   |
| Literature reading                                | 2.5 hours   |
| Implementation (model, dataloader, training loop) | 4 hours     |
| Debugging (channel mismatch, PyTorch 2.6 issues)  | 2 hours     |
| Training runs                                     | 1.5 hours   |
| Evaluation & visualisation                        | 1 hour      |
| Report writing                                    | 2 hours     |
| Total                                             | ~14.5 hours |

Machine: Google Colab free tier

GPU: NVIDIA T4, 16 GB VRAM, 1 GPU

Training time: approximately 3 minutes per epoch, approximately 90 minutes total (30 epochs, 1,000 samples)

Resource constraints: Colab free tier session limit (~4-5 hours) and storage constraints prevented training on the full 8.8 GB dataset. A 1,000-sample subset was used as a principled compromise, explicitly documented in Section 3.6.

## 8. References

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